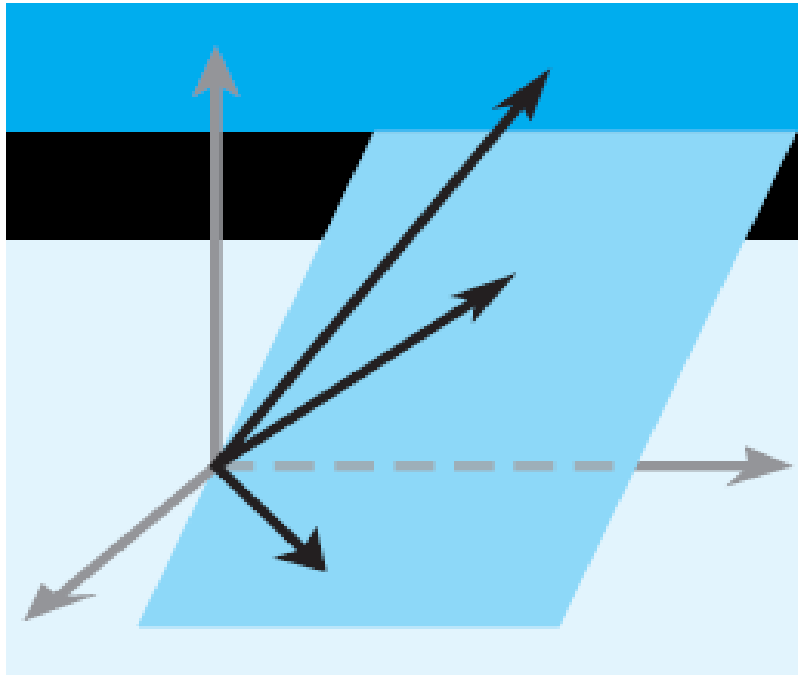


# Elementary Linear Algebra

## Chapter 6



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# Chapter 6 Inner Product Spaces

- **6.1 Inner Products**
- 6.2 Angle and Orthogonality in Inner Product Spaces
- 6.3 Gram-Schmidt Process; QR-Decomposition
- 6.4 Best Approximation; Least Squares
- 6.5 Least Squares Fitting to Data
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# Section 6.1 Inner Products

## Section 3.2 Dot Products

### DEFINITION 4

If  $\mathbf{u} = (u_1, u_2, \dots, u_n)$  and  $\mathbf{v} = (v_1, v_2, \dots, v_n)$  are vectors in  $R^n$ , then the *dot product* (also called the *Euclidean inner product*) of  $\mathbf{u}$  and  $\mathbf{v}$  is denoted by  $\mathbf{u} \cdot \mathbf{v}$  and is defined by

$$\mathbf{u} \cdot \mathbf{v} = u_1 v_1 + u_2 v_2 + \dots + u_n v_n = \mathbf{u} \mathbf{v}^T$$

|          |          |
|----------|----------|
| $u_1$    | $v_1$    |
| $u_2$    | $v_2$    |
| $\vdots$ | $\vdots$ |
| $u_n$    | $v_n$    |

$$\mathbf{u} \bullet \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

# Section 6.1 Inner Products

**DEFINITION 1** An *inner product* on a real vector space  $V$  is a function that associates a real number  $\langle \mathbf{u}, \mathbf{v} \rangle$  with each pair of vectors in  $V$  in such a way that the following axioms are satisfied for all vectors  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  in  $V$  and all scalars  $k$ .

1.  $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$  [Symmetry axiom]
2.  $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$  [Additivity axiom]
3.  $\langle k\mathbf{u}, \mathbf{v} \rangle = k\langle \mathbf{u}, \mathbf{v} \rangle$  [Homogeneity axiom]
4.  $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$  and  $\langle \mathbf{v}, \mathbf{v} \rangle = 0$  if and only if  $\mathbf{v} = \mathbf{0}$  [Positivity axiom]

A real vector space with an inner product is called a *real inner product space*.

# Section 6.1 Inner Products

## DEFINITION 2

If  $V$  is a real inner product space, then the *norm* (or *length*) of a vector  $\mathbf{v}$  in  $V$  is denoted by  $\|\mathbf{v}\|$  and is defined by

$$\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$$

and the *distance* between two vectors is denoted by  $d(\mathbf{u}, \mathbf{v})$  and is defined by

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \sqrt{\langle \mathbf{u} - \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle}$$

A vector of norm 1 is called a *unit vector*.

# Section 6.1 Inner Products

## THEOREM 6.1.1

*If  $\mathbf{u}$  and  $\mathbf{v}$  are vectors in a real inner product space  $V$ , and if  $k$  is a scalar, then:*

*(a)  $\|\mathbf{v}\| \geq 0$  with equality if and only if  $\mathbf{v} = \mathbf{0}$ .*

*(b)  $\|k\mathbf{v}\| = |k| \|\mathbf{v}\|$ .*

*(c)  $d(\mathbf{u}, \mathbf{v}) = d(\mathbf{v}, \mathbf{u})$ .*

*(d)  $d(\mathbf{u}, \mathbf{v}) \geq 0$  with equality if and only if  $\mathbf{u} = \mathbf{v}$ .*

# Section 6.1 Inner Products

## ► EXAMPLE 1 Weighted Euclidean Inner Product

Let  $\mathbf{u} = (u_1, u_2)$  and  $\mathbf{v} = (v_1, v_2)$  be vectors in  $\mathbb{R}^2$ . Verify that the weighted Euclidean inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2 \quad (3)$$

satisfies the four inner product axioms.

### **Solution**

**Axiom 1:** Interchanging  $\mathbf{u}$  and  $\mathbf{v}$  in Formula 3 does not change the sum on the right side, so  $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$ .

**Axiom 2:** If  $\mathbf{w} = (w_1, w_2)$ , then

$$\begin{aligned} \langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle &= 3(u_1 + v_1)w_1 + 2(u_2 + v_2)w_2 \\ &= 3(u_1w_1 + v_1w_1) + 2(u_2w_2 + v_2w_2) \\ &= (3u_1w_1 + 2u_2w_2) + (3v_1w_1 + 2v_2w_2) \\ &= \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle \end{aligned}$$

# Section 6.1 Inner Products

**Axiom 3:**  $\langle k\mathbf{u}, \mathbf{v} \rangle = 3(ku_1)v_1 + 2(ku_2)v_2$   
 $= k(3u_1v_1 + 2u_2v_2)$   
 $= k\langle \mathbf{u}, \mathbf{v} \rangle$

**Axiom 4:**  $\langle \mathbf{v}, \mathbf{v} \rangle = 3(v_1v_1) + 2(v_2v_2) = 3v_1^2 + 2v_2^2 \geq 0$  with equality if and only if  $v_1 = v_2 = 0$ , that is, if and only if  $\mathbf{v} = \mathbf{0}$ .



# Section 6.1 Inner Products

## ►EXAMPLE 2 Using a Weighted Euclidean Inner Product

It is important to keep in mind that norm and distance depend on the inner product being used. If the inner product is changed, then the norms and distances between vectors also change. For example, for the vectors  $\mathbf{u} = (1, 0)$  and  $\mathbf{v} = (0, 1)$  in  $\mathcal{R}^2$  with the Euclidean inner product we have

$$\|\mathbf{u}\| = \sqrt{1^2 + 0^2} = 1$$

and

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\| = \|(1, -1)\| = \sqrt{1^2 + (-1)^2} = \sqrt{2}$$

but if we change to the weighted Euclidean inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2$$

► **EXAMPLE 2** Using a Weighted Euclidean Inner Product

$$\| \mathbf{u} \| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = [3(1)(1) + 2(0)(0)]^{1/2} = \sqrt{3}$$

and

$$\begin{aligned} d(\mathbf{u}, \mathbf{v}) &= \| \mathbf{u} - \mathbf{v} \| = \langle (1, -1), (1, -1) \rangle^{1/2} \\ &= [3(1)(1) + 2(-1)(-1)]^{1/2} = \sqrt{5} \end{aligned}$$

# Section 6.1 Inner Products

## ► EXAMPLE 3 Unusual Unit Circles in $R^2$

- (a) Sketch the unit circle in an  $xy$ -coordinate system in  $R^2$  using the Euclidean inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = u_1v_1 + u_2v_2$ .
- (b) Sketch the unit circle in an  $xy$ -coordinate system in  $R^2$  using the weighted Euclidean inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = \frac{1}{9}u_1v_1 + \frac{1}{4}u_2v_2$ .

**Solution (a)** If  $\mathbf{u} = (x, y)$ , then  $\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{x^2 + y^2}$ , so the equation of the unit circle is  $\sqrt{x^2 + y^2} = 1$ , or on squaring both sides,

$$x^2 + y^2 = 1$$

As expected, the graph of this equation is a circle of radius 1 centered at the origin (Figure 6.1.1a).

**Solution (b)** If  $\mathbf{u} = (x, y)$ , then  $\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{\frac{1}{9}x^2 + \frac{1}{4}y^2}$ , so the equation

# Section 6.1 Inner Products

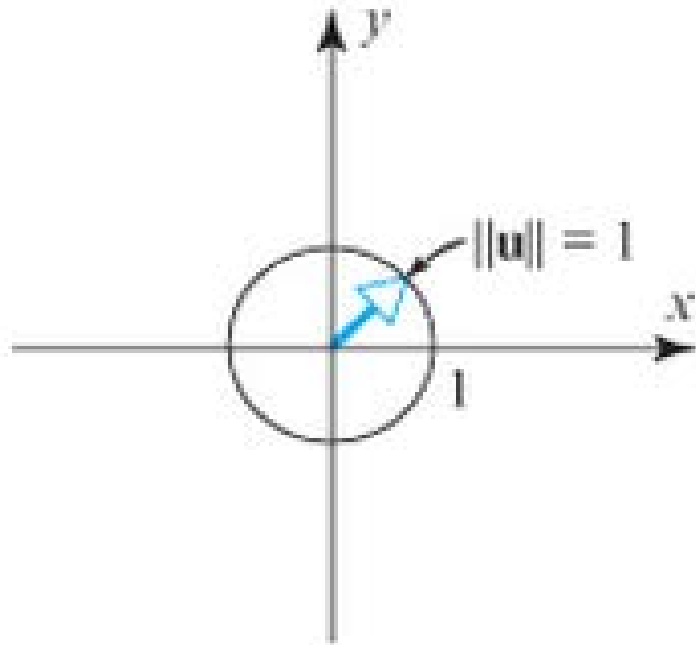
**Solution (b)** If  $\mathbf{u} = (x, y)$ , then  $\|\mathbf{u}\| = \langle \mathbf{u}, \mathbf{u} \rangle^{1/2} = \sqrt{\frac{1}{9}x^2 + \frac{1}{4}y^2}$ , so the equation

of the unit circle is  $\sqrt{\frac{1}{9}x^2 + \frac{1}{4}y^2} = 1$ , or on squaring both sides,

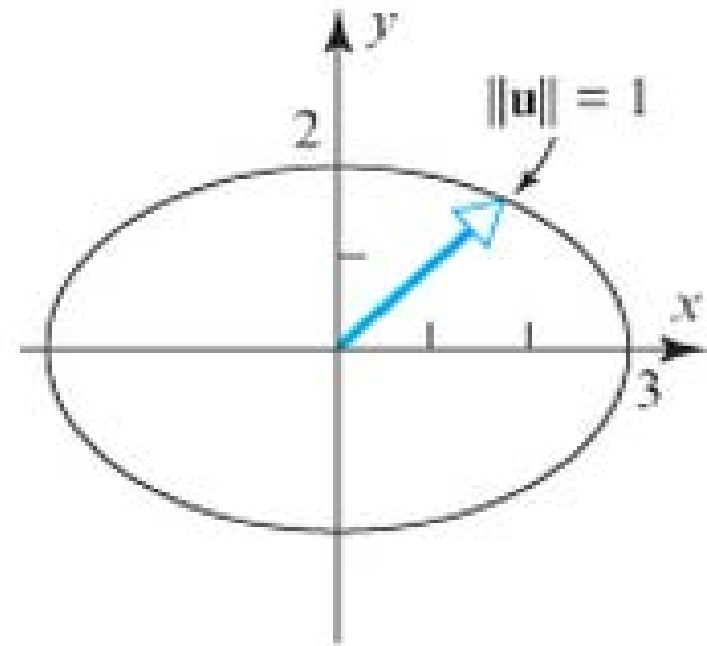
$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

The graph of this equation is the ellipse shown in Figure [6.1.1b](#).

# Section 6.1 Inner Products



(a) The unit circle using the standard Euclidean inner product.



(b) The unit circle using a weighted Euclidean inner product.

► **Figure 6.1.1**

# Section 6.1 Inner Products

## Matrices Generating Weighted Euclidean Inner Products

Inner Products on  $\mathbb{R}^n$  generated by  $A$

$$\begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix} \quad \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$\langle u, v \rangle = Au \bullet Av$$

$$= (Av)^T Au$$

$$= v^T A^T Au$$

# Section 6.1 Inner Products

## ► EXAMPLE 4 Matrices Generating Weighted Euclidean Inner Products

The standard Euclidean and weighted Euclidean inner products are special cases of matrix inner products. The standard Euclidean inner product on  $\mathbf{R}^n$  is generated by the  $n \times n$  identity matrix, since setting  $A = I$  in Formula (4) yields

$$\langle \mathbf{u}, \mathbf{v} \rangle = I\mathbf{u} \cdot I\mathbf{v} = \mathbf{u} \cdot \mathbf{v}$$

and the weighted Euclidean inner product

$$\langle \mathbf{u}, \mathbf{v} \rangle = w_1 u_1 v_1 + w_2 u_2 v_2 + \cdots + w_n u_n v_n$$

(6)

# Section 6.1 Inner Products

## ► EXAMPLE 4 **Matrices Generating Weighted Euclidean Inner Products**

is generated by the matrix

$$A = \begin{bmatrix} \sqrt{w_1} & 0 & 0 & \cdots & 0 \\ 0 & \sqrt{w_2} & 0 & \cdots & 0 \\ \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & 0 & \cdots & \sqrt{w_n} \end{bmatrix} \quad (7)$$

This can be seen by observing that  $A^T A$  is the  $n \times n$  diagonal matrix whose diagonal entries are the weights  $w_1, w_2, \dots, w_n$  and then observing that (5) simplifies to (6) when  $A$  is the matrix in Formula (7).

# Section 6.1 Inner Products

## ►EXAMPLE 5 Example [1](#) Revisited

The weighted Euclidean inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = 3u_1v_1 + 2u_2v_2$  discussed in Example [1](#) is the inner product on  $\mathbb{R}^2$  generated by

$$A = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & \sqrt{2} \end{bmatrix}$$

# Section 6.1 Inner Products

## ► **EXAMPLE 6** An Inner Product on $M_{nn}$

If  $U$  and  $V$  are  $n \times n$  matrices, then the formula

$$\langle U, V \rangle = \text{trace}(U^T V) \quad (8)$$

defines an inner product on the vector space  $M_{nn}$  (see Definition 8 of Section 1.3 for a definition of trace). This can be proved by confirming that the four inner product space axioms are satisfied, but we can see why this is so by computing (8) for the  $2 \times 2$  matrices

$$\langle \mathbf{u}, \mathbf{v} \rangle = \text{tr}(U^T V) = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

This yields

$$U = \begin{bmatrix} u_1 & u_2 \\ u_3 & u_4 \end{bmatrix} \quad \text{and} \quad V = \begin{bmatrix} v_1 & v_2 \\ v_3 & v_4 \end{bmatrix}$$

# Section 6.1 Inner Products

## ► EXAMPLE 6 An Inner Product on $M_{nn}$

which is just the dot product of the corresponding entries in the two matrices. For example, if

$$U = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \quad \text{and} \quad V = \begin{bmatrix} -1 & 0 \\ 3 & 2 \end{bmatrix}$$

Then

$$\langle U, V \rangle = 1(-1) + 2(0) + 3(3) + 4(2) = 16$$

The norm of a matrix  $U$  relative to this inner product is

$$\|U\| = \langle U, U \rangle^{1/2} = \sqrt{u_1^2 + u_2^2 + u_3^2 + u_4^2}$$

# Section 6.1 Inner Products

► **EXAMPLE 7** The Standard Inner Product on  $P_n$

**Option**

If

$$\mathbf{p} = a_0 + a_1x + \cdots + a_nx^n \quad \text{and} \quad \mathbf{q} = b_0 + b_1x + \cdots + b_nx^n$$

are polynomials in  $P_n$ , then the following formula defines an inner product on  $P_n$  (verify) that we will call the *standard inner product* on this space:

$$\langle \mathbf{p}, \mathbf{q} \rangle = a_0b_0 + a_1b_1 + \cdots + a_nb_n \quad (9)$$

The norm of a polynomial  $\mathbf{p}$  relative to this inner product is

$$\| \mathbf{p} \| = \sqrt{\langle \mathbf{p}, \mathbf{p} \rangle} = \sqrt{a_0^2 + a_1^2 + \cdots + a_n^2}$$

# Section 6.1 Inner Products

## ► EXAMPLE 8 The Evaluation Inner Product on $P_n$ **Option**

If

$$\mathbf{p} = p(x) = a_0 + a_1x + \cdots + a_nx^n \quad \text{and} \quad \mathbf{q} = q(x) = b_0 + b_1x + \cdots + b_nx^n$$

are polynomials in  $P_n$ , and if  $x_0, x_1, \dots, x_n$  are distinct real numbers (called *sample points*), then the formula

$$\langle \mathbf{p}, \mathbf{q} \rangle = p(x_0)q(x_0) + p(x_1)q(x_1) + \cdots + p(x_n)q(x_n) \quad (10)$$

defines an inner product on  $P_n$  called the *evaluation inner product* at  $x_0, x_1, \dots, x_n$ .

Algebraically, this can be viewed as the dot product in  $R^n$  of the  $n$ -tuples

$$(p(x_0), p(x_1), \dots, p(x_n)) \quad \text{and} \quad (q(x_0), q(x_1), \dots, q(x_n))$$

and hence the first three inner product axioms follow from properties of the dot product. The fourth inner product axiom follows from the fact that

$$\langle \mathbf{p}, \mathbf{p} \rangle = [p(x_0)]^2 + [p(x_1)]^2 + \cdots + [p(x_n)]^2 \geq 0$$

# Section 6.1 Inner Products

**Option**

with equality holding if and only if

$$p(x_0) = p(x_1) = \cdots = p(x_n) = 0$$

But a nonzero polynomial of degree  $n$  or less can have at most  $n$  distinct roots, so it must be that  $\mathbf{p} = \mathbf{0}$ , which proves that the fourth inner product axiom holds.

The norm of a polynomial  $\mathbf{p}$  relative to the evaluation inner product is

$$\|\mathbf{p}\| = \sqrt{\langle \mathbf{p}, \mathbf{p} \rangle} = \sqrt{[p(x_0)]^2 + [p(x_1)]^2 + \cdots + [p(x_n)]^2} \quad (11)$$

# Section 6.1 Inner Products

**Option**

## ► **EXAMPLE 9** Working with the Evaluation Inner Product

Let  $P_2$  have the evaluation inner product at the points

$$x_0 = -2, \quad x_1 = 0, \quad \text{and} \quad x_2 = 2$$

Compute  $\langle \mathbf{p}, \mathbf{q} \rangle$  and  $\|\mathbf{p}\|$  for the polynomials  $\mathbf{p} = p(x) = x^2$  and  $\mathbf{q} = q(x) = 1 + x$ .

**Solution**

It follows from [10](#) and [11](#) that

$$\begin{aligned}\langle \mathbf{p}, \mathbf{q} \rangle &= p(-2)q(-2) + p(0)q(0) + p(2)q(2) = (4)(-1) + (0)(1) + (4)(3) = 8 \\ \|\mathbf{p}\| &= \sqrt{[p(x_0)]^2 + [p(x_1)]^2 + [p(x_2)]^2} = \sqrt{[p(-2)]^2 + [p(0)]^2 + [p(2)]^2} \\ &= \sqrt{4^2 + 0^2 + 4^2} = \sqrt{32} = 4\sqrt{2}\end{aligned}$$

# Section 6.1 Inner Products

## ► EXAMPLE 10 An Integral Inner Product on $C[a, b]$

Let  $\mathbf{f} = f(x)$  and  $\mathbf{g} = g(x)$  be two functions in  $C[a, b]$  and define

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_a^b f(x) g(x) dx \quad (12)$$

We will show that this formula defines an inner product on  $C[a, b]$  by verifying the four inner product axioms for functions,  $\mathbf{f} = f(x)$ ,  $\mathbf{g} = g(x)$ , and  $\mathbf{h} = h(x)$  in  $C[a, b]$ :

1.

$$\langle \mathbf{f}, \mathbf{g} \rangle = \int_a^b f(x) g(x) dx = \int_a^b g(x) f(x) dx = \langle \mathbf{g}, \mathbf{f} \rangle$$

2.

$$\begin{aligned} \langle \mathbf{f} + \mathbf{g}, \mathbf{h} \rangle &= \int_a^b (f(x) + g(x)) h(x) dx \\ &= \int_a^b f(x) h(x) dx + \int_a^b g(x) h(x) dx \\ &= \langle \mathbf{f}, \mathbf{h} \rangle + \langle \mathbf{g}, \mathbf{h} \rangle \end{aligned}$$

# Section 6.1 Inner Products

3. 
$$\langle k\mathbf{f}, \mathbf{g} \rangle = \int_a^b k f(x) g(x) dx = k \int_a^b f(x) g(x) dx = k \langle \mathbf{f}, \mathbf{g} \rangle$$

which proves that Axiom 3 holds.

4. If  $\mathbf{f} = f(\mathbf{x})$  is any function in  $\mathbf{C}[a, b]$ , then

$$\langle \mathbf{f}, \mathbf{f} \rangle = \int_a^b f^2(x) dx \geq 0 \quad (13)$$

since  $f^2(x) \geq 0$  for all  $\mathbf{x}$  in the interval  $[a, b]$ . Moreover, because  $f$  is continuous on  $[a, b]$ , the equality in Formula 13 holds if and only if the function  $f$  is identically zero on  $[a, b]$ , that is, if and only if  $\mathbf{f} = \mathbf{0}$ ; and this proves that Axiom 4 holds.

# Section 6.1 Inner Products

## ►EXAMPLE 11 Norm of a Vector in $C[a, b]$

If  $C[a, b]$  has the inner product that was defined in Example [10](#), then the norm of a function  $\mathbf{f} = f(x)$  relative to this inner product is

$$\|\mathbf{f}\| = \langle \mathbf{f}, \mathbf{f} \rangle^{1/2} = \sqrt{\int_a^b f^2(x) dx} \quad (14)$$

and the unit sphere in this space consists of all functions  $\mathbf{f}$  in  $C[a, b]$  that satisfy the equation

$$\int_a^b f^2(x) dx = 1$$

# Section 6.1 Inner Products

**THEOREM 6.1.2** *If  $\mathbf{u}$ ,  $\mathbf{v}$ , and  $\mathbf{w}$  are vectors in a real inner product space  $V$ , and if  $k$  is a scalar, then:*

$$(a) \quad \langle \mathbf{0}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{0} \rangle = 0$$

$$(b) \quad \langle \mathbf{u}, \mathbf{v} + \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle + \langle \mathbf{u}, \mathbf{w} \rangle$$

$$(c) \quad \langle \mathbf{u}, \mathbf{v} - \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{w} \rangle$$

$$(d) \quad \langle \mathbf{u} - \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle - \langle \mathbf{v}, \mathbf{w} \rangle$$

$$(e) \quad k \langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{u}, k\mathbf{v} \rangle$$

# Section 6.1 Inner Products

## ► EXAMPLE 12 Calculating with Inner Products

$$\begin{aligned}\langle \mathbf{u} - 2\mathbf{v}, 3\mathbf{u} + 4\mathbf{v} \rangle &= \langle \mathbf{u}, 3\mathbf{u} + 4\mathbf{v} \rangle - \langle 2\mathbf{v}, 3\mathbf{u} + 4\mathbf{v} \rangle \\&= \langle \mathbf{u}, 3\mathbf{u} \rangle + \langle \mathbf{u}, 4\mathbf{v} \rangle - \langle 2\mathbf{v}, 3\mathbf{u} \rangle - \langle 2\mathbf{v}, 4\mathbf{v} \rangle \\&= 3\langle \mathbf{u}, \mathbf{u} \rangle + 4\langle \mathbf{u}, \mathbf{v} \rangle - 6\langle \mathbf{v}, \mathbf{u} \rangle - 8\langle \mathbf{v}, \mathbf{v} \rangle \\&= 3\|\mathbf{u}\|^2 + 4\langle \mathbf{u}, \mathbf{v} \rangle - 6\langle \mathbf{u}, \mathbf{v} \rangle - 8\|\mathbf{v}\|^2 \\&= 3\|\mathbf{u}\|^2 - 2\langle \mathbf{u}, \mathbf{v} \rangle - 8\|\mathbf{v}\|^2\end{aligned}$$

# Homework

## Sec 6.1

### Practice for Quiz on 3/24

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# 27 (a)      Page 379

# 31      Page 379

# Theorem 6.1.2 proof      Page 340

